Cloud Engineer Roadmap

From zero to job-ready, using free study resources

My step-by-step plan: command line, CompTIA A+, AWS, Terraform and a portfolio project, with a budget home lab built alongside. I started with no command line, programming or networking experience, so this plan starts from the beginning.

Last updated October 2026. Exam versions and links were checked on that date. Every resource is listed with a link in my Training Resources document.

1. The plan at a glance

Goal: become a cloud engineer who can work remotely, starting with IT support and building up through certifications and hands-on projects.

Phase	When	Focus	Finish line
1	Days 1-28	Command line, Linux basics, Git	I can work in a terminal without notes
2	Days 29-112	CompTIA A+ (two exams), plus networking and Python basics alongside	A+ certified
3	Days 113-154	AWS Cloud Practitioner	CLF-C02 passed
4	Days 155-252	Python automation, networking, Docker, Cloud Resume Challenge	Live cloud project on GitHub
5	Days 253-364	AWS Solutions Architect Associate, Terraform Associate	Two more certifications
6	Day 365+	Kubernetes, CI/CD, monitoring; apply for cloud roles	Cloud support or junior cloud engineer role

Pace: this plan is counted in **training days**, not calendar days or weeks. A training day equals one training session, typically 0.5 to 2 hours of focused study. Every 7 training days follow the rhythm in Section 3: five study days, a lab day and a review day. I do not skip the checkpoints. (1 to 1.5 hours on a normal study day and 2 to 3 hours on lab and review days. That's roughly 10 hours per 7-day block, so the total is about 500 hours across all 364 training days.)

Experience alongside the study: labs and portfolio projects in every phase give me real, demonstrable work to point to as I earn each certification.

2. Which certification first

Starting with CompTIA A+

A+ is two exams, 220-1201 (Core 1) and 220-1202 (Core 2), and both are required. This is the current V15 series, released March 25, 2025, with retirement estimated around September 2028. Both exams must come from the same series. Professor Messer's free course covers every exam objective: Core 1 is 63 videos (10 hr 11 min) and Core 2 is 74 videos (13 hr 41 min). CompTIA raised prices on June 1, 2026, so I confirm the current price on the CompTIA A+ page before paying.

Why A+ before AWS

- It teaches what I am missing: hardware, operating systems, networking, security and troubleshooting. Cloud work sits on top of these.
- Employers screen help desk and IT support candidates for A+. It is the key that opens the first door.
- The full course is free.
- The AWS Cloud Practitioner is a good second certification and a cheap one. If an employer cares more about cloud than A+, I can flip the order.

Exams in this plan

Exam	Fee	Notes
CompTIA A+ Core 1 and Core 2	Check comptia.org	Two separate exams
AWS Certified Cloud Practitioner (CLF-C02)	$100	65 questions, 90 minutes, valid 3 years
AWS Certified Solutions Architect Associate (SAA-C03)	$150 ($75 with voucher)	65 questions, 130 minutes, valid 3 years
HashiCorp Terraform Associate (004)	$70.50	60 minutes, online proctored, valid 2 years, no free retake

AWS gives 50% off the next exam after passing any AWS exam, found in the AWS Certification account.

3. How I study

Rhythm for each 7-day block

- **Training days 1-5, about 1 hour each:** watch videos or read, with notes in my own words.
- **Training day 6, 2.5 hours:** hands-on lab. I type every command myself and never just watch.
- **Training day 7, 2.5 hours:** review flashcards and quizzes, update my notes, plan the next block.

Rules that make it work

- My notes are kept as text in this GitHub repository, so there is a visible record of my learning.
- I use spaced-repetition flashcards (Anki, free) for commands, port numbers, AWS services and acronyms.
- Practice exams are saved for the last 14 training days of each certification. I schedule the real exam only after scoring 85% or higher on several practice tests in a row.
- If a lab fails, I read the error message out loud, search the exact text and write down the fix. Troubleshooting is the job.

Tools

- My laptop is my first lab, running Ubuntu. I check RAM with `free -h` (8 GB runs two small VMs, 16 GB is comfortable) and confirm virtualization is on in the BIOS.
- VirtualBox for virtual machines, Visual Studio Code as my editor, and a free GitHub account.
- Later, a small used mini PC and a managed switch extend the lab (see Phase 4 and the Homelab section).

4. Phase 1: Command line and Linux basics (days 1-28)

Goal: get comfortable in a terminal. Almost every cloud tool is driven from the command line.

Days	What I do	Resources
1-7	Open a terminal. Learn pwd, ls, cd, mkdir, touch, cp, mv, rm, cat, man. Finish the Linux Journey "Grasshopper" section. Bandit levels 0-5. Start an Anki deck of commands.	Linux Journey OverTheWire Bandit Ubuntu command line tutorial
8-14	Linux Journey "Novice": file permissions, users and groups, links. Bandit levels 6-12. Read Part 1 of the free Linux Command Line book. Lab: build a folder tree, change permissions with chmod, create a user.	Linux Journey The Linux Command Line (free book)
15-21	Processes, package management with apt, text editors (nano), pipes and redirection, grep. Bandit levels 13-20. Lab: install and remove software with apt, find a message in a log with grep, write a five-line shell script.	Linux Journey OverTheWire Bandit
22-28	Learn Git: install git, read chapters 1-2 of Pro Git, finish the "Introduction to GitHub" course. Create a repository for my study notes and commit to it. Watch the intro video of the A+ course.	Pro Git book GitHub Skills

Checkpoint

- [ ] Create, move, copy and delete files and folders from the terminal.
- [ ] Read a permissions line such as -rwxr-xr-- and change it with chmod.
- [ ] Search a file with grep and chain two commands with a pipe.
- [ ] Install a package with apt and write a short bash script that runs.
- [ ] Commit my notes and push them to GitHub.

5. Phase 2: CompTIA A+ (days 29-112)

Goal: pass both A+ exams. Each training week is about 7 hours of A+ and 3 hours of Python and networking basics. I start CS50P now: it is free, has 10 weeks of lectures and problem sets, and needs only a web browser. I do one CS50P week every 7-10 training days.

Days	A+ study	Alongside
29-42	Core 1 from the first video: laptop hardware, motherboards, memory, storage, power. Hands-on: on Linux, run lscpu, lsblk and free -h to identify my own hardware.	CS50P weeks 0-1 (functions, variables, conditionals)
43-56	Core 1 networking and mobile devices: ports, protocols, wireless, Ethernet, small-office networks. Hands-on: ip addr, ping, traceroute and nslookup.	CS50P weeks 2-3 (loops, exceptions)
57-70	Core 1 virtualization, cloud concepts, and hardware and network troubleshooting. Build my first virtual machine in VirtualBox. Daily pop quizzes, then sit Core 1 at the end of this block.	Subnetting: the IPv4 Addressing videos in the free Network+ course, plus Practical Networking. CS50P week 4
71-84	Core 2 operating systems: Windows, macOS and Linux tools, installation, command-line utilities.	CS50P weeks 5-6 (unit tests, file I/O)
85-98	Core 2 security: authentication, malware, wireless security, social engineering, software troubleshooting.	CS50P weeks 7-8 (regular expressions, object-oriented programming)
99-105	Core 2 operational procedures, documentation, safety and professionalism. Review all weak areas.	CS50P week 9
106-112	Daily pop quizzes and full review. Sit Core 2.	Start the CS50P final project

Free A+ resources: Core 1 videos, Core 1 pop quizzes, Core 2 videos, Core 2 pop quizzes.

Checkpoint

- [ ] A+ Core 1 and Core 2 passed.
- [ ] I can explain what an IP address, subnet mask, gateway, DNS and DHCP do.
- [ ] I can write a short Python program with a loop and a function.

6. Phase 3: AWS Cloud Practitioner (days 113-154)

Goal: pass CLF-C02 and earn my first cloud credential. The exam is 65 questions in 90 minutes and costs $100. My A+ and Linux work should make it much easier.

Days	What I do	Resources
113-119	Set up AWS safely first: create an account, turn on multi-factor authentication for the root user, create a $5 budget alert in Billing, create an admin IAM user and stop using root. Then start the exam prep plan and read the exam guide.	Skill Builder exam prep plan Exam guide AWS Free Tier
120-133	Core services: IAM, EC2, S3, VPC basics, RDS, Lambda, CloudFront, Route 53. Labs: launch a free-tier EC2 instance, connect with SSH, install a web server with apt, then terminate it. Create an S3 bucket and host a static web page. Shut down every lab afterward.	Skill Builder exam prep plan A free CLF-C02 video course (confirm it covers CLF-C02)
134-140	Security and compliance (shared responsibility, IAM best practices), pricing, billing tools, support plans, the Well-Architected pillars.	Exam guide
141-147	Official practice question set, official pretest and official practice exam. Fix every weak area.	Skill Builder exam prep plan
148-154	Sit the exam. Then claim the 50% voucher for my next AWS exam.	Cloud Practitioner page

> **Cost safety**
>
> Cloud bills can surprise beginners. Free-tier terms change, so I read the current terms on the Free Tier page, keep my budget alert on, and terminate resources after every lab.

7. Phase 4: Build the cloud toolkit (days 155-252)

Goal: turn knowledge into things I built. This phase produces the portfolio project.

> **Home lab step: add a used mini PC (days 134-175)**
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> My laptop and VirtualBox cover Phases 1-3. From here I start networking, Docker and infrastructure as code, so I add an always-on machine to experiment on without risking my laptop.
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> **Models to look for:** Lenovo ThinkCentre M720q, HP EliteDesk 800 G4 Mini, or Dell OptiPlex 3060 Micro. **Specs:** at least 8 GB RAM and a 256 GB SSD, with 16 GB or more better. **Checks before buying:** virtualization support (VT-x), no BIOS password, healthy SSD. **Run on it:** Ubuntu Server or Proxmox, both free.

Days	What I do	Resources
155-182	Python for automation. Finish the CS50P final project. Then write scripts with the AWS SDK for Python (boto3): one that lists my S3 buckets, one that stops EC2 instances by tag. Daily small exercises.	boto3 quickstart Exercism Python track

Days	What I do	Resources
183-210	Networking in depth. Network+ course sections 1-3 (concepts, implementation, operations). Hands-on: in AWS, build a custom VPC with public and private subnets, a route table, an internet gateway and security groups, and draw the diagram. At home, add a managed switch and practice VLANs between my laptop and mini PC. The Network+ exam is optional.	Network+ N10-009 free course Practical Networking
211-224	Docker. Work through the Get Started guide. Containerize my CS50P project and run the container on an EC2 instance and on my home lab mini PC.	Docker Get Started
225-252	Cloud Resume Challenge (AWS version). Host my resume as a static site on S3 and CloudFront, add a visitor counter with Lambda, API Gateway and DynamoDB, and deploy updates with GitHub Actions. Publish everything on GitHub.	Cloud Resume Challenge GitHub Actions quickstart

Checkpoint

- [ ] My resume site is live and the code is on GitHub with a clear README.
- [ ] I can explain my VPC design and why the subnets and security groups are set up the way they are.

8. Phase 5: Solutions Architect and Terraform (days 253-364)

Goal: earn the certification most cloud postings recognize, then add infrastructure as code. AWS recommends about a year of hands-on experience for the Solutions Architect Associate, which my labs and Phase 4 projects supply.

Days	What I do	Resources
253-266	Read the SAA-C03 exam guide. The four domains are secure, resilient, high-performing and cost-optimized architectures. Follow the Skill Builder 4-step plan.	SAA exam page Exam guide Skill Builder plan
267-294	Compute, storage and networking in depth: EC2 purchasing options, Auto Scaling, load balancers, S3 storage classes, EBS and EFS, VPC peering, Route 53, CloudFront. A lab for each topic.	Skill Builder plan
295-308	Databases, serverless, decoupling (SQS, SNS), and security (KMS, IAM policies). Re-read my weakest notes.	Skill Builder plan
309-322	Practice exams: the official practice question set, pretest and practice exam. Review every wrong answer and write why.	Skill Builder plan
323-329	Sit the Solutions Architect Associate ($150, or $75 with my voucher).	AWS Certification account
330-357	Terraform. Do the six AWS get-started tutorials, then the fundamentals (CLI, configuration language, modules, state). Rebuild my resume challenge infrastructure in Terraform, and practice Terraform against local VMs on my home lab. Then the Associate (004) prep tutorials.	Terraform AWS tutorials Associate (004) prep
358-364	Sit the Terraform Associate (004): one hour, online proctored, $70.50.	HashiCorp certification page

9. Year 2 and beyond

With A+, Cloud Practitioner, Solutions Architect Associate, Terraform Associate and a live portfolio, I will be ready to apply for cloud roles. Year 2 is for widening my skills.

- **Kubernetes:** the official tutorials. It has moved from nice-to-have to expected in many DevOps and platform postings.
- **CI/CD and monitoring:** build pipelines with GitHub Actions and set up basic monitoring dashboards.
- **Deeper Linux:** systemd, networking tools, bash scripting, logs and troubleshooting.
- **Optional security track:** Security+ is the common next step if I lean toward cloud security (free course).

Roles I am working toward, in order

- Cloud support associate, cloud operations, or junior DevOps (often the first cloud title).
- Junior or mid-level cloud engineer.
- Cloud engineer with Terraform and Kubernetes, then SRE or platform engineering later.

10. If I fall behind

- Missing a few days is fine, since I only count the days I actually train. Missing the checkpoint is not: I repeat the lab until I can do it without notes.
- If a certification takes twice as long as planned, I keep the hands-on work going and move only the exam date.
- I do not buy multiple courses for the same topic. One good video course plus labs beats three half-finished ones.
- I get unstuck with the Discord and forum links on the course sites, and by searching the exact error text.

11. Home lab

Everything here is optional until Phase 4. I start with the laptop I already have, then add the rest as training reaches networking, Docker and Terraform.

Device	Recommended	Why
Laptop (already owned)	Ubuntu laptop with VirtualBox	First lab for Linux, A+ practice and my first VMs. 8 GB of RAM or more.
Mini PC (main lab server)	Lenovo ThinkCentre M720q, HP EliteDesk 800 G4 Mini, or Dell OptiPlex 3060 Micro, bought used	Runs Ubuntu Server or Proxmox, VMs, Docker and Terraform practice, and can stay on all the time. 16 GB of RAM or more.
RAM upgrade	A second matching DDR4 laptop-style (SODIMM) stick	Two sticks run faster and allow more VMs.
Storage	SSD of 256 GB or more	Holds VM disks and install images.
Managed switch	TP-Link TL-SG108E or Netgear GS308E	Practice VLANs and port mirroring in the networking phase.
Cables and USB drive	A few short Ethernet cables and a 16 GB+ flash drive	Connect the lab and make install media.
Operating systems	Ubuntu Server and Proxmox VE (both free)	Ubuntu Server for Linux practice, Proxmox for managing VMs.

Order to get things

- **Now:** my laptop and VirtualBox.
- **Days 134-175:** the used mini PC, plus a second RAM stick and an SSD if it needs them.
- **Around day 183**, when networking starts: the managed switch and a few Ethernet cables.
- **With my first server:** a USB drive for install media.

What I will practice on it

- Linux server setup and SSH
- VMs and Docker containers
- DHCP, DNS and subnets
- VLANs

- Terraform against local VMs

Skipping for now

- Old rack servers: loud, power-hungry, and bigger than I need.
- Building a PC from parts: a complete used mini PC is cheaper and works out of the box.
- A cloud lab at home: the AWS free tier covers the cloud side.